Tiny ECG Classification on ESP32

A 73.9% smaller full-int8 CNN with 95.1% test accuracy

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FROM-SCRATCH CNN / TFLITE MICRO

Project Goal

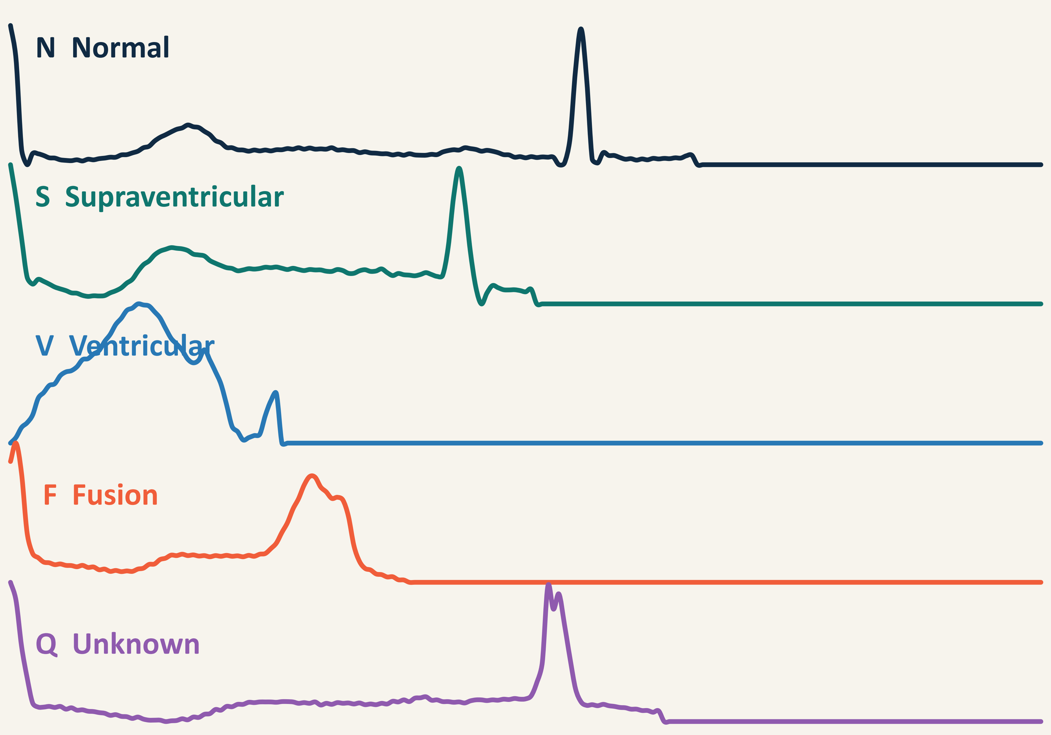
Develop a **1D CNN from scratch** in NumPy to accurately classify different heartbeat types, apply **INT8 quantization** to reduce the model size, and optimize it for efficient deployment on an **ESP32**.

95.12%	73.9%	52 ms
INT8 TEST ACCURACY	SMALLER MODEL	ESP32 INFERENCE / BEAT

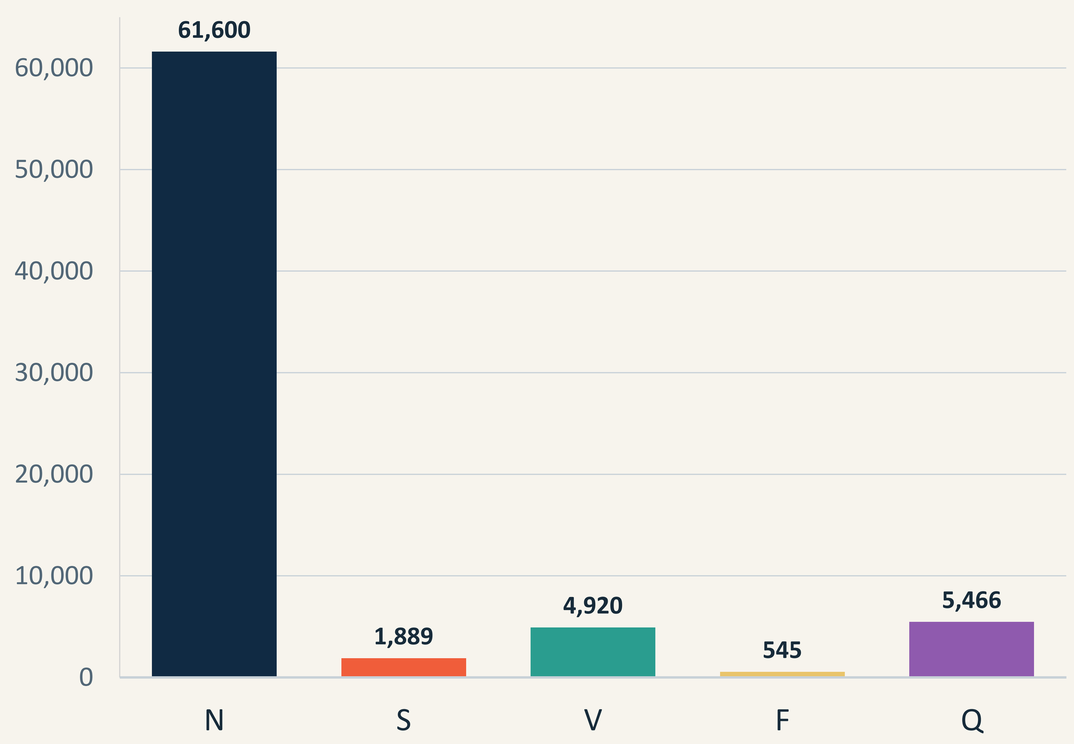
1. Heartbeat data are highly imbalanced

Five heartbeat types must be separated from 187-point ECG segments. Normal beats dominate training, so overall accuracy alone can hide minority-class failure.

Representative normalized beats

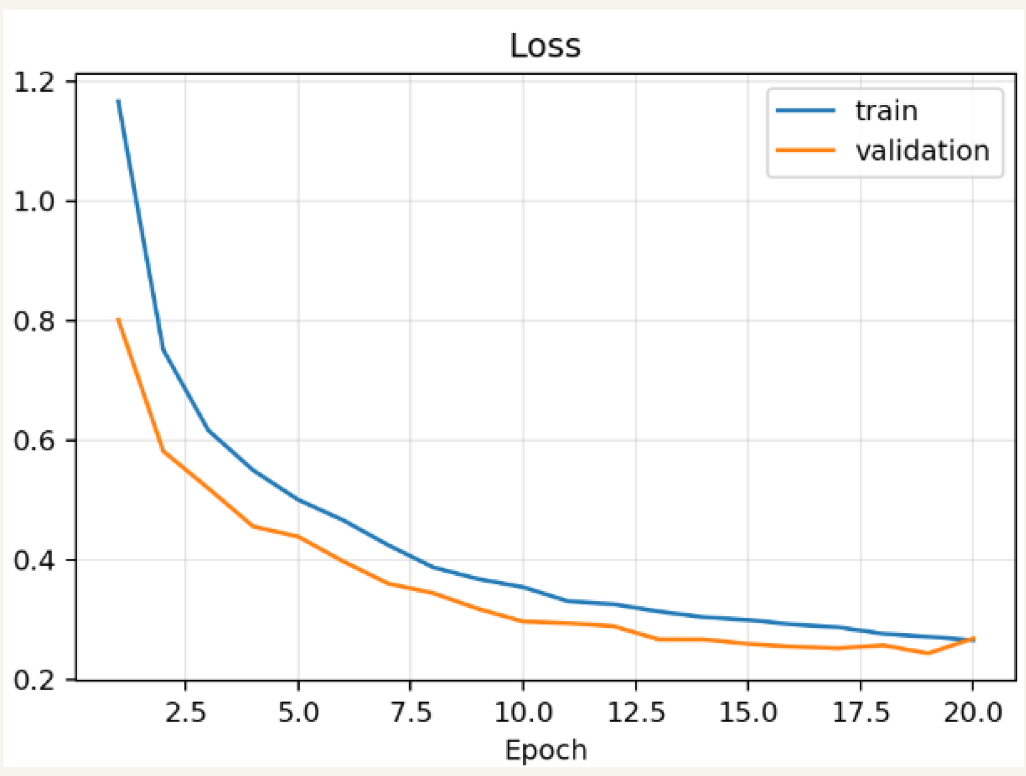


Training class distribution



Class F has 113x fewer examples than class N.

Stable convergence without clear overfitting



2. Compact CNN, explicit imbalance controls

The complete forward pass, backward pass, weighted loss, and Adam updates are implemented in NumPy. Keras is used only as an export bridge.

11,173 trainable parameters

187 x 1 ECG beat
normalized heartbeat segment

↓

Three convolution blocks
16 (k=7) -> 32 (k=5) -> 64 (k=3)

↓

Global average pooling
small representation, no large flatten layer

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Dense 32 + dropout
compact nonlinear classifier

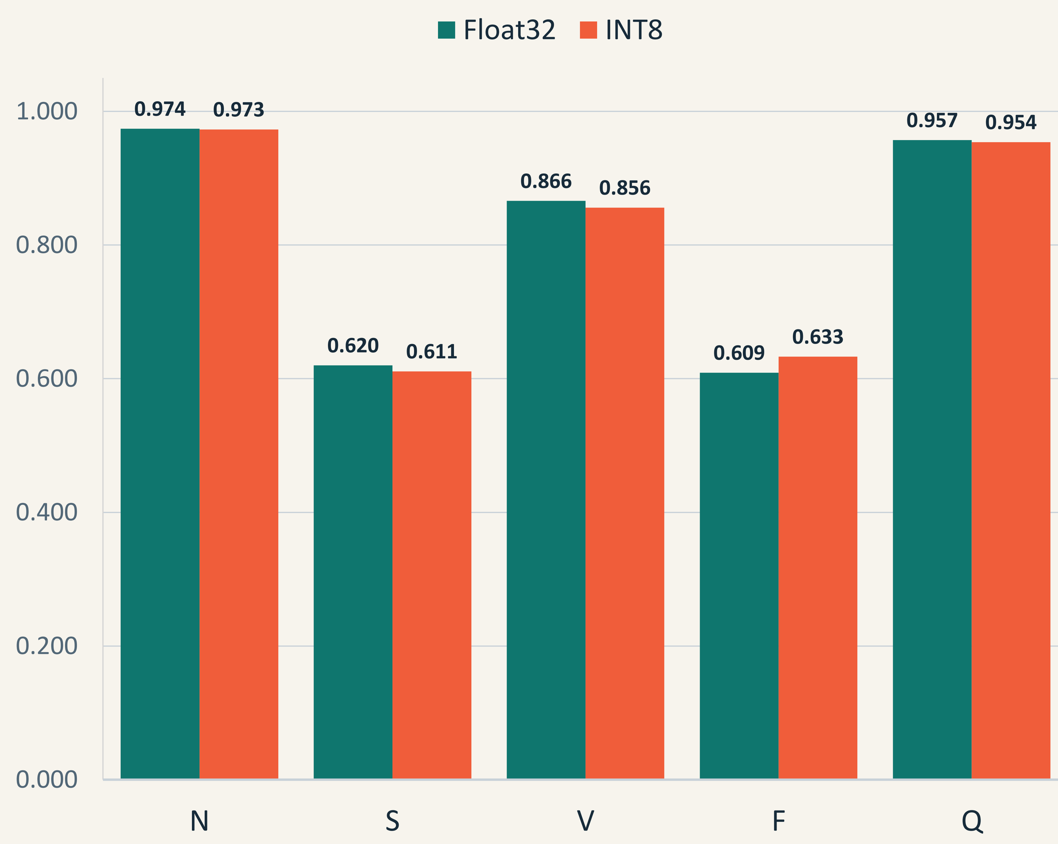
↓

Five-class softmax
N / S / V / F / Q

Three complementary strategies

1. **Capped class-weighted loss**
   Inverse-frequency penalties, capped at 4 to avoid unstable minority gradients.
2. **Mild rare-class augmentation**
   Noise, amplitude scaling, baseline shift, and small time shifts preserve beat shape.
3. **Sampling controls**
   Natural shuffle, inverse-frequency weighted sampling, or balanced mini-batches.

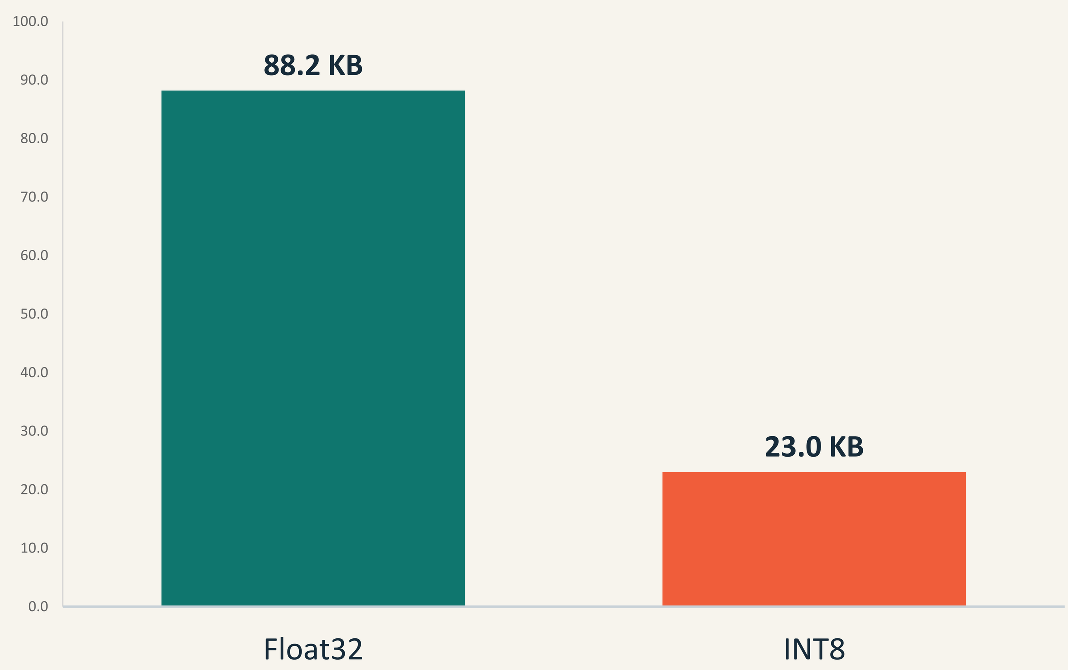
Quantization preserves per-class F1



3. Full-int8 compression keeps model quality

Representative-data calibration quantizes weights, activations, inputs, and outputs to int8 for TensorFlow Lite Micro.

Model Size



-73.9% size | -0.14 percentage points accuracy

INT8 normalized confusion matrix

	N	S	V	F	Q
N	0.96	0.01	0.02	0.01	0.00
S	0.37	0.56	0.04	0.01	0.01
V	0.02	0.00	0.95	0.02	0.00
F	0.10	0.00	0.09	0.81	0.00
Q	0.02	0.00	0.01	0.00	0.97

Predicted class

Class S remains hardest: 37% are confused with normal beats.

Deployment path

Float32 CNN 90,361 B → **Full INT8 TFLite** 23,576 B → **ESP32 + TFLM** 52 ms / beat

80 KiB tensor arena | full-int8 input/output | real MIT-BIH beats

Prototype scope

Validated on stored test beats. Live ADC acquisition, heartbeat segmentation, patient-wise clinical validation, and confidence handling remain future work.

TAKEAWAY **Full-int8 quantization is the deployment sweet spot: 23 KiB, 95.1% accuracy, and real-time ESP32 inference.**